

Copper rod at 50 Hz

Hybrid A- Φ finite-element solver, tree-cotree gauge. A slender copper conductor driven by a 1 V terminal voltage, validated against the exact axisymmetric solution.

regression_tests/02_Ansys_Cylinder_50Hz · 240,990 unknowns ·
backward error 4.8e-21

0. How to read these figures

Every figure here comes from one solve, on one mesh — the Netgen-optimised mesh in §1. "Netgen-optimised" is a property of that mesh, not a variant of any individual plot: there is no unoptimised figure here to compare against. What Netgen changed, and by how much, is in §2.1.

Only one pair differs in *processing* rather than in what it shows: §4.2 applies one smoothing pass, and those are the only two images with a red banner burned into them.

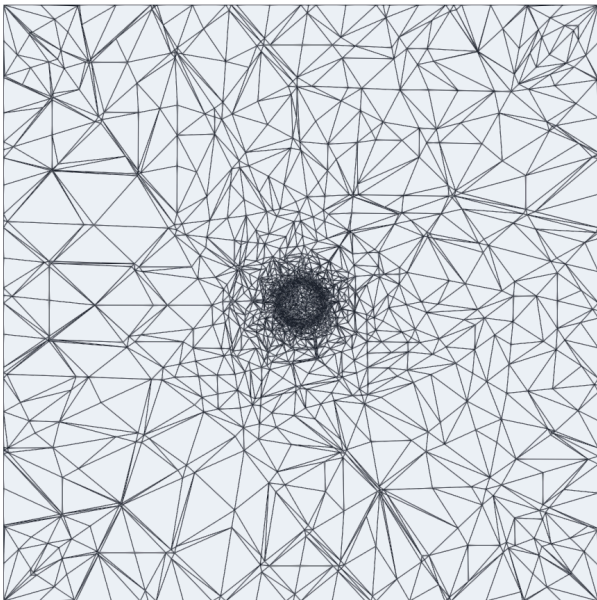
SECTION	WHAT THE FIGURE IS
§2	cross-section mesh, whole domain and the wire alone
§3	$\text{Re}(\Phi)$ and $ \Phi $ on the wire's lateral surface
§4.1	nodal $ B $, unsmoothed — the solver's output
§4.2	the same data, one smoothing pass — not solver output
§5.1	per-node $ E $, whole domain, log scale
§5.1.1	where the nodal field is not to be trusted (no figure)
§5.2	nodal $ E $ in the wire, auto range beside pinned range
§5.3	J as coloured vectors, longitudinal cut

1. The case

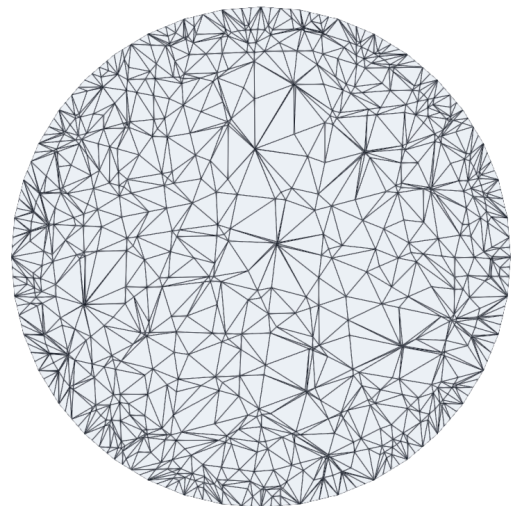
Conductor	copper, $\sigma = 5.8e7 \text{ S/m}$, $a = 1.5 \text{ mm}$, $L = 40 \text{ mm}$, a 40-gon
Domain	40 mm cube of air, $\mathbf{n} \times \mathbf{A} = \mathbf{0}$ on the outer boundary
Excitation	1.0 V top face, 0.0 V bottom face, 50 Hz
Skin depth	$\delta = 9.346 \text{ mm}$, so $a/\delta = 0.1605$ — no skin effect
Regime	$\omega L/R = 0.0725$, resistance dominated, so Φ is readable
Formulation	first-order Whitney edge $\mathbf{A}$, second-order P2 nodal Φ
Mesh	18,994 nodes, 101,873 tets, Netgen-optimised , 0.25 mm at the surface, 0.4 mm in the core
Solve	240,990 unknowns, 503 s, 4.03 GB

a/δ and $\omega L/R$ are the same parameter, both scaling as ωa^2 . A case cannot show a strong skin effect and a readable potential at once; this one takes the resistive branch.

2. Mesh



Whole domain. Graded from 0.25 mm at the conductor to 6 mm in the far air.



The wire alone. Roughly 6 elements across the radius.

The element budget sits where the field is: **6.7 %** inside $r = 1 \text{ mm}$, **84.8 %** in the band $1.0\text{--}2.0 \text{ mm}$, and **3.8 %** beyond $r = 3 \text{ mm}$.

2.1 What Netgen optimisation bought

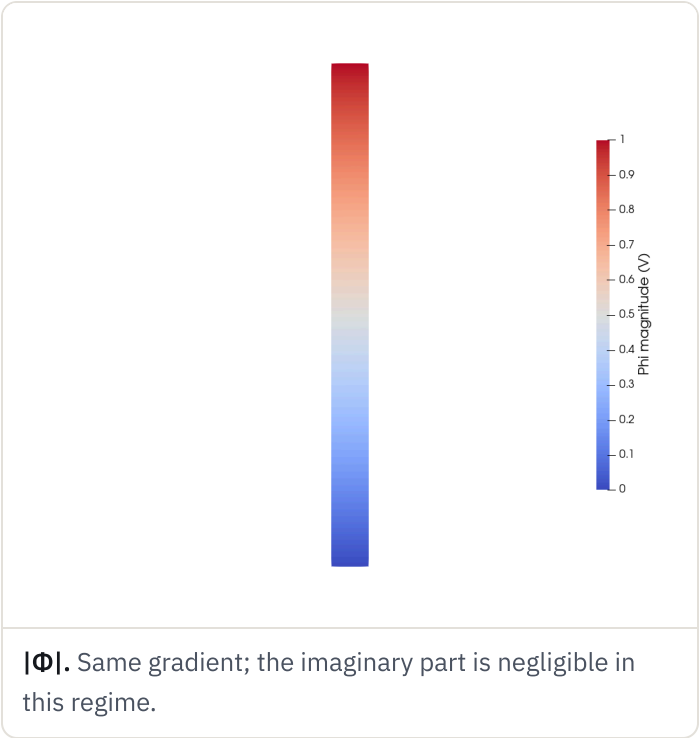
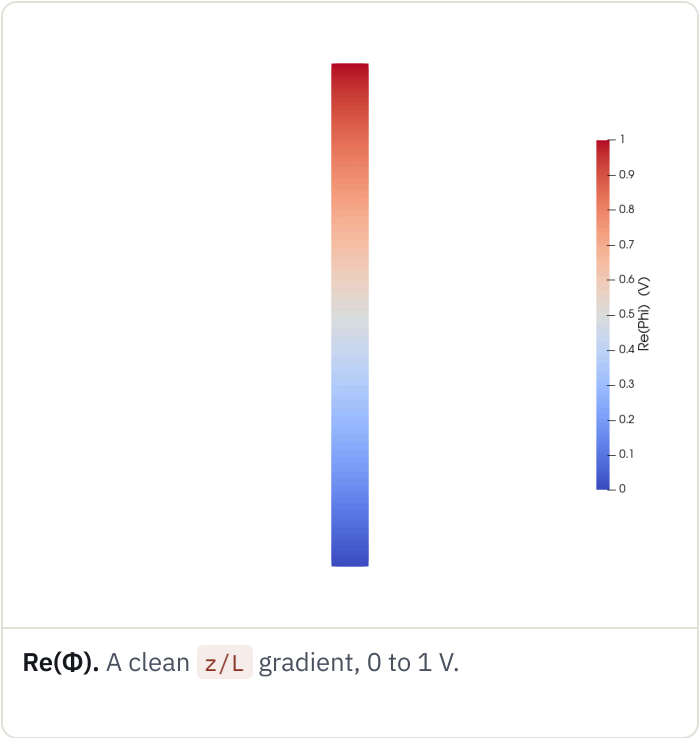
It changes element *shape* at fixed element size. Measured against the same mesh generated without it — same geometry file, same sizes, only the flag changed:

	WITHOUT	WITH NETGEN	
tets	105,715	101,873	CombineImprove merges elements
unknowns	248,294	240,990	
factorise	681 s, 4.56 GB	503 s, 4.03 GB	26 % faster, 12 % less memory
scatter at the peak	0.0181	0.0138	−24 %
scatter near the core	0.0309	0.0210	−32 %
azimuthal purity	0.9996	0.9999	

Better-shaped elements give the sparse ordering less fill-in, so the quality gain pays for itself twice. For comparison, the last *refinement* step bought 0.0215 → 0.0181 for 1.6× the unknowns and 2.8× the time. Netgen bought 0.0181 → 0.0138 for nothing.

3. Potential

Φ over the lateral surface of the wire, seen side on — not a cross-section, because the potential is driven along the axis.

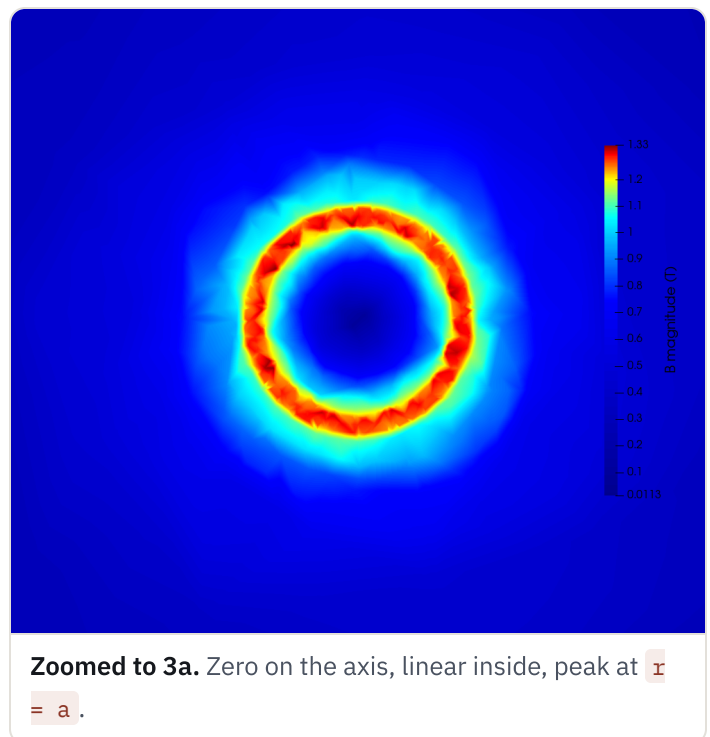
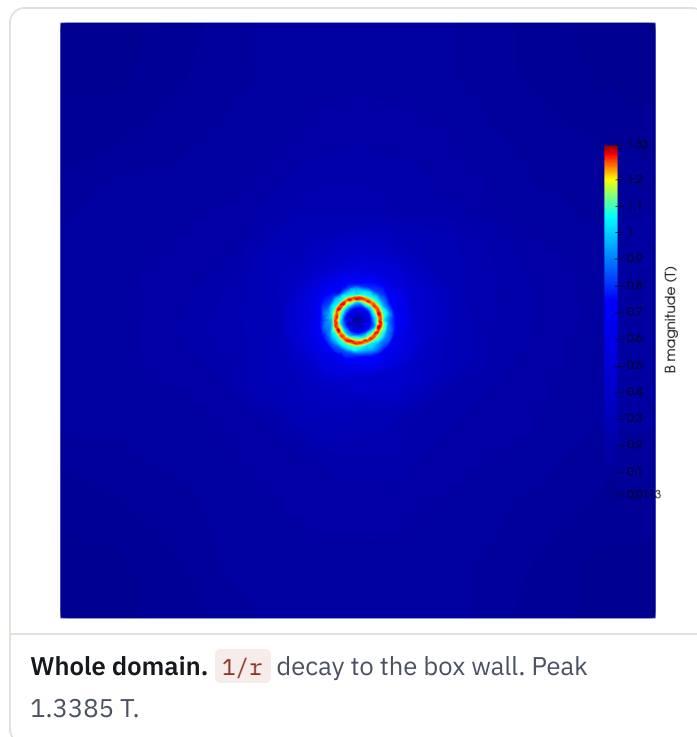


Against the exact z/L over all 95,149 wire nodes: worst deviation **8.26e-04**, mean **2.51e-04**. Φ is gauge dependent — a different spanning tree shifts it by $-j\omega\psi$, almost purely imaginary. **E**, **B**, **H** and **J** are not.

4. Magnetic flux density

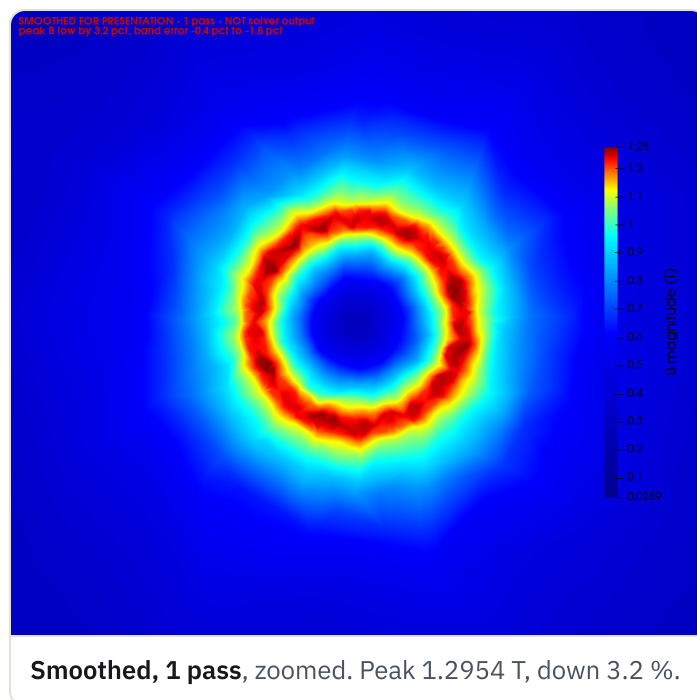
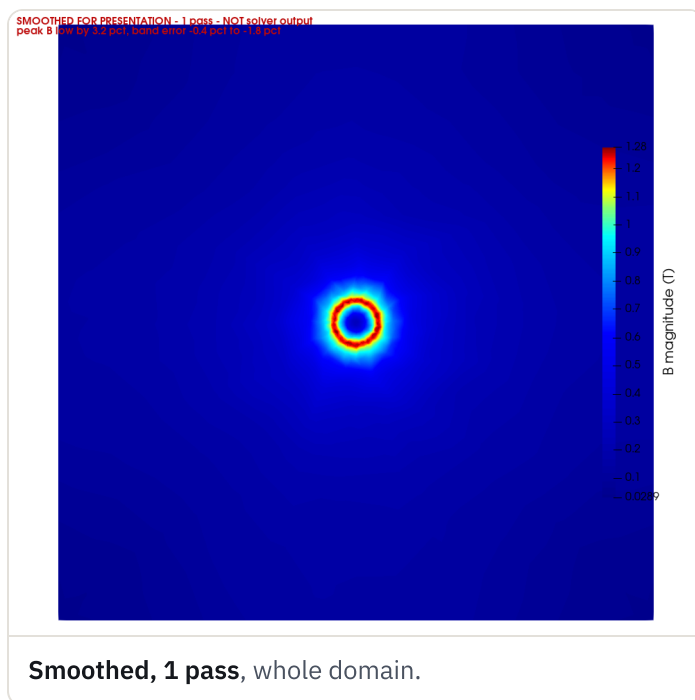
4.1 The solver's output

Nodal **|B|**, mid-length cross-section, **no smoothing**. Linear colour scale — a log ramp compresses the $1/r$ decay into the top few colours and hides the ring entirely. These are the figures to read numbers off.



4.2 The same data with one smoothing pass

The only difference from §4.1 is one point-to-cell-to-point averaging round trip. Same solve, same mesh, same colour map. Commercial tools smooth plotted fields by default, which is the leading explanation for why their **|B|** cross-sections look cleaner, so these exist to compare like with like. **They are not the solver's output**, and each carries a red banner saying so.

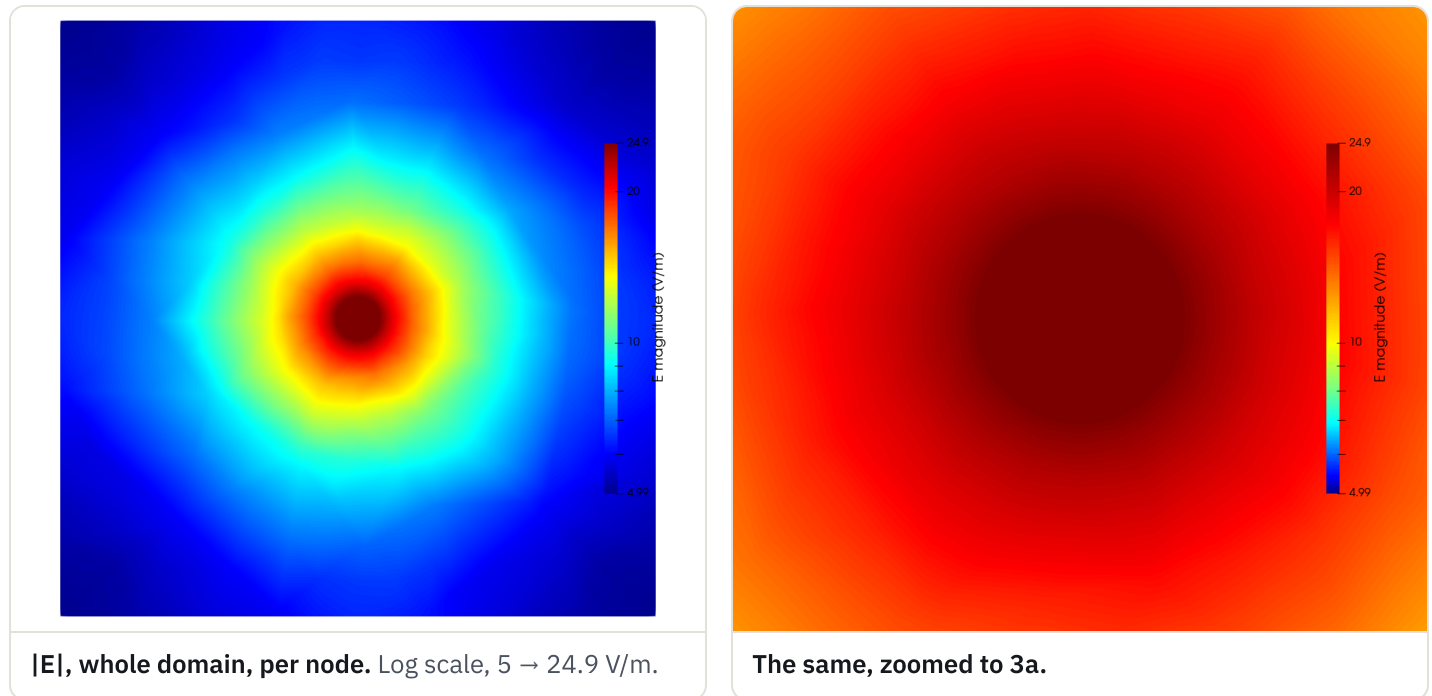


	UNSMOOTHED (§4.1)	1 PASS (§4.2)	2 PASSES	4 PASSES
scatter, 1.20–1.49 mm	0.0296	0.0183	0.0179	0.0225
error, 1.20–1.49 mm	–0.39 %	–1.84 %	–2.43 %	–3.48 %
displayed peak (T)	1.3385	1.2954	1.2716	1.2387
peak change	—	–3.2 %	–5.0 %	–7.5 %

38 % less scatter for 3.2 % of the peak — a better trade than on the pre-Netgen mesh, where it cost 6.2 % for 14 %. The bias still grows while the scatter stops improving past one pass. Smoothing is not a route to a better answer; it is the control needed to compare like with like if the reference picture is itself smoothed.

5. Electric field

5.1 Whole domain, per node



The scale is logarithmic, **5 → 24.9 V/m**. The conductor is at the **top** of that range, which is correct: **E** is largest inside the copper and decays outward.

5.1.1 Where the nodal field is not to be trusted

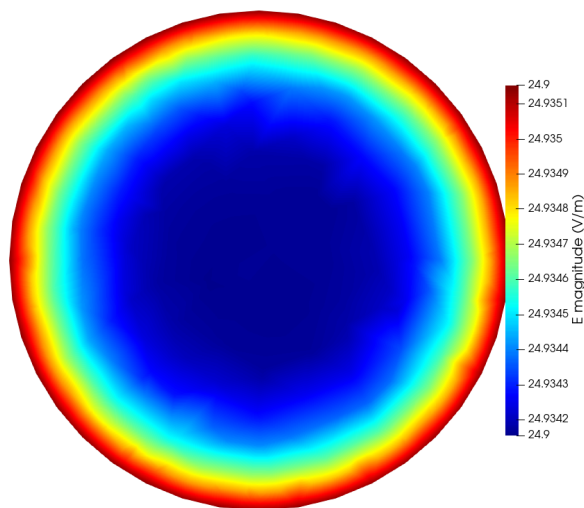
The post-processor also writes a per-cell **E**, one constant value per tetrahedron. It is not plotted here — it renders as a mosaic and reads badly — but it is the reference the nodal field is checked against, because it is exact at the material interface where the nodal field is not. Mean **|E|** by radius, nodal against per cell:

BAND (MM)	NODAL	PER CELL	DIFFERENCE
0.0 – 1.5	24.934	24.934	0.00 %
1.5 – 2.0	29.972	37.882	–20.88 %
2.0 – 3.0	29.866	29.775	+0.31 %
3.0 – 5.0	21.507	21.493	+0.06 %
5.0 – 8.0	14.807	14.690	+0.80 %

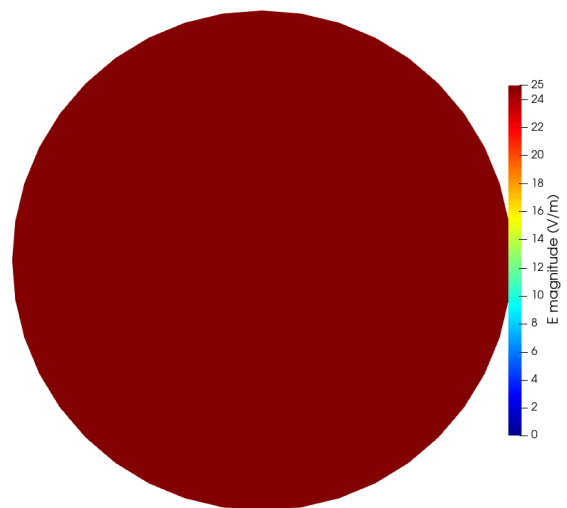
Inside the conductor the two agree exactly, and from $r = 2 \text{ mm}$ outward to better than 1 %. **Exactly one band differs — the one immediately outside the wire.** E 's *normal* component jumps at the material interface, so no single nodal value is correct there. The post-processor resolves the ambiguity by taking the higher-conductivity side, so every interface node carries the copper value, 24.93 V/m — but the air just outside genuinely carries 37.88 V/m, because a radial component appears there that the purely axial field inside does not have. $|E|$ jumps **up** crossing into the air, so forcing the copper value onto those nodes makes that first band read **21 % low**, not high.

45,200 of 140,532 nodes (32.2 %) are flagged as interface nodes, all at $r = 1.4954\text{--}1.5000 \text{ mm}$. Use the per-cell figure when the value at the surface matters; use the nodal one when the overall picture does.

5.2 Inside the wire — and why the colours disagree with §5.1



Auto colour range. Range 24.9342 → 24.9351 V/m.



Range pinned 0–25 V/m. The same data.

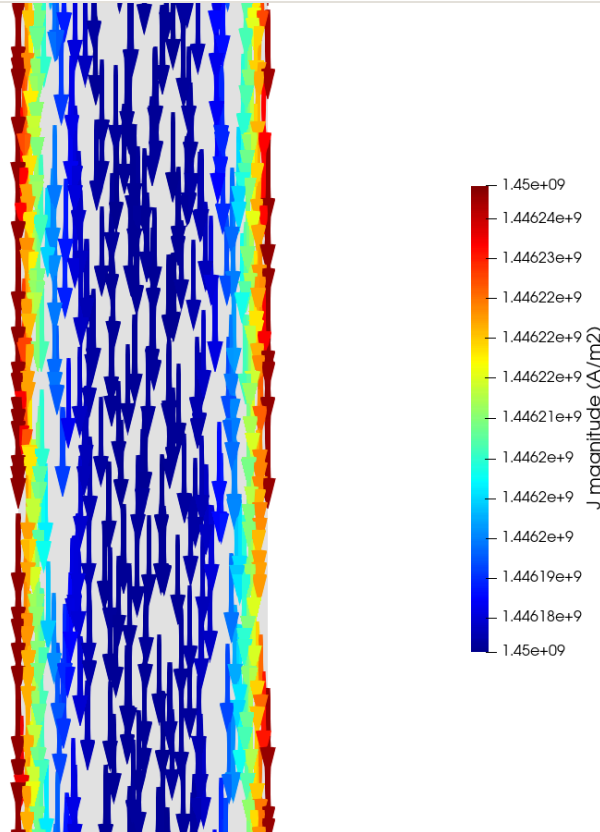
This answers why the centre reads red in one figure and blue in the other. Both show the same number, about 24.935 V/m, against colour ranges that differ by a factor of roughly 20,000.

In §5.1 the range is $5 \rightarrow 24.9 \text{ V/m}$ across the whole box, so the conductor sits at the top and renders dark red. In the left figure above, ParaView rescaled to the wire's *own* range — **24.9342 → 24.9351 V/m**, a total span of $9\text{e-}04 \text{ V/m}$, or **0.004 %** of the value. Auto-

rescaling blew that up to the full colour map, producing a red ring and blue core that look like a strong radial gradient and are in fact noise at the fifth significant figure.

The right figure pins the range to $0-25 \text{ V/m}$ and the same data renders as a uniform disc — the physical truth. With $a/\delta = 0.16$ there is no skin effect and $E = V/L$ uniformly. The exact value is 25.0 V/m ; we are **0.26 %** low, consistent with the 40-gon being 0.41 % smaller in area than the circle it approximates.

5.3 Current density



$\mathbf{J} = \sigma \mathbf{E}$, on a longitudinal slice. On a mid-length cut every arrow points at the viewer and nothing is visible.

6. Validation

Nothing below fits a free parameter. The current follows from R_{dc} under the 1 V drive: $I = 10207.3 \text{ A}$, $B(a) = 1.3610 \text{ T}$.

R	9.796995e-05 Ω against 9.796863e-05 exact — 1.3e-05
L from $\text{Im}(Z)/\omega$	22.60 nH
$J(\theta)/J(a)$	0.999968 against the Bessel 0.999959
Φ vs exact z/L	worst 8.26e-04, mean 2.51e-04
B direction at the peak	azimuthal component 0.9999 of the total — 0.01 % off
B inside the conductor	every band within 0.5 %

BAND (MM)	MEAN, T	EXACT, T	ERROR	AZIMUTHAL SCATTER
0.30 – 0.60	0.4167	0.4163	+0.24 %	0.0477
0.60 – 0.90	0.6914	0.6884	+0.40 %	0.0309
0.90 – 1.20	0.9906	0.9929	–0.22 %	0.0210
1.20 – 1.49	1.2734	1.2722	+0.10 %	0.0138
1.49 – 2.00	1.2380	1.2633	–1.99 %	0.0307
2.00 – 3.00	0.8776	0.8869	–1.15 %	0.0559
3.00 – 5.00	0.5490	0.5601	–2.01 %	0.0768
5.00 – 8.00	0.3267	0.3349	–2.37 %	0.0868

Linear rise inside, $1/r$ outside, correct absolute scale, over a 25× span in radius. Inside the conductor every band is within 0.5 %.

7. Known limitations

- **Azimuthal scatter in B of 1.4–3.1 % near the conductor.** The problem is axisymmetric, so this is error. It is $O(h)$ element noise from B being constant per tetrahedron — the lowest-order quantity in the formulation.
- **Nodal B reads low in the far field — the figures in §4, not the numbers in §6.** §4 plots the nodal array; §6 is computed from the per-cell one. Against the exact solution at each sample's own radius: **0.30–1.49 mm** nodal –0.4 to –1.5 % against per-cell +0.1 to +0.4 %; then **1.49–2.00** –4.29 vs –2.01 %, **3.00–5.00** –6.15 vs –2.12 %, **5.00–8.00** **–7.50** vs –2.19 %. Inside the conductor and at the peak both are within 1.5 %, so the 1.3385 T peak is sound; beyond the wire the nodal colours under-read. Likely cause, reasoned rather than

separately measured: the nodal value is a volume-weighted average over each node's element patch, the mesh coarsens rapidly outward, and the patch is therefore dominated by the larger outer elements where $|B| \sim 1/r$ is smaller. Unlike E , B has **no** material-interface problem here, because $\mu_r = 1$ on both sides — the band that is E 's worst is B 's best. A magnetic material would change that.

- **A coherent polygon harmonic is now visible, at 0.31 %.** Fourier analysis at $r = a$ finds a peak at $m = 40$, the polygon order. No earlier mesh showed one — it was buried under element-quality noise, and Netgen removed enough of that to expose it. At 0.31 % against 1.4 % total scatter it is not what the eye sees, but it means raising N becomes worthwhile once quality noise falls further.
- **Far-field accuracy was traded for near-field resolution.** Beyond $r = 3 \text{ mm}$ the error is $\sim 2 \%$ against $\sim 1 \%$ on a uniformly-graded mesh. That band holds 3.8 % of the elements and does not affect R , L or the conductor fields.
- **Tried and rejected, all measured:** superconvergent patch recovery (made B more than twice as rough), plot smoothing (costs peak amplitude for modest gain), raising the polygon order alone (no effect on scatter, and slivers above $N = 48$), coarsening the far field further (a few per cent of the mesh to reclaim).
- **Uniform refinement is reaching its limit.** lc_{skin} $0.7 \rightarrow 0.35 \rightarrow 0.25$ gave peak-band scatter $0.029 \rightarrow 0.0215 \rightarrow 0.0181$, ratios $\times 0.74$ and $\times 0.84$ where $O(h)$ allows $\times 0.50$ and $\times 0.71$. Netgen then took it to **0.0138 for free** — a larger gain than the last refinement step bought. Element *shape* was the cheaper lever all along.